1.3 Mechanics

This topic covers rectilinear motion, forces, energy and power. It may be studied using applications that relate to mechanics such as sports.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

1	be able to use the equations for uniformly accelerated motion in one dimension: $s = \frac{(u+v)t}{2}$ $v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
2	be able to draw and interpret displacement-time, velocity-time and acceleration-time graphs
3	know the physical quantities derived from the slopes and areas of displacement-time, velocity-time and acceleration-time graphs, including cases of non-uniform acceleration and understand how to use the quantities
4	understand scalar and vector quantities and know examples of each type of quantity and recognise vector notation
5	be able to resolve a vector into two components at right angles to each other by drawing and by calculation
6	be able to find the resultant of two coplanar vectors at any angle to each other by drawing, and at right angles to each other by calculation
7	understand how to make use of the independence of vertical and horizontal motion of a projectile moving freely under gravity
8	be able to draw and interpret free-body force diagrams to represent forces on a particle or on an extended but rigid body using the concept of *centre of gravity* of an extended body
9	be able to use the equation $\sum F = ma$, and understand how to use this equation in situations where *m* is constant (Newton's second law of motion), including Newton's first law of motion where $a = 0$, objects at rest or travelling at constant velocity *Use of the term 'terminal velocity' is expected.*
10	be able to use the equations for gravitational field strength $g = \frac{F}{m}$ and weight $W = mg$
11	**CORE PRACTICAL 1: Determine the acceleration of a freely-falling object**
12	know and understand Newton's third law of motion and know the properties of pairs of forces in an interaction between two bodies

14	know the principle of conservation of linear momentum, understand how to relate this to Newton's laws of motion and understand how to apply this to problems in one dimension
15	be able to use the equation for the moment of a force, moment of force = Fx where x is the perpendicular distance between the line of action of the force and the axis of rotation
16	be able to use the concept of centre of gravity of an extended body and apply the principle of moments to an extended body in equilibrium
17	be able to use the equation for work $\Delta W = F\Delta s$, including calculations when the force is not along the line of motion
18	be able to use the equation $E_k = \frac{1}{2}mv^2$ for the kinetic energy of a body
19	be able to use the equation $\Delta E_{grav} = mg\Delta h$ for the difference in gravitational potential energy near the Earth's surface
20	know, and understand how to apply, the principle of conservation of energy including use of work done, gravitational potential energy and kinetic energy
21	be able to use the equations relating power, time and energy transferred or work done $P = \frac{E}{t}$ and $P = \frac{W}{t}$
22	be able to use the equations $\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$ and $\text{efficiency} = \frac{\text{useful power output}}{\text{total power input}}$